

Iterations (Loops)

1. Ask for a number N and print all numbers from 1 to N using. [\[ans\]](#)
2. Calculate the sum of all natural numbers up to N using a while loop.
3. Calculate the factorial of a number N using a while loop.
[\[ans\]](#) ($N = N \times (N-1) \times \dots \times 1$).
4. Take an integer as input and print its reverse (e.g., 12345 -> 54321) using a while loop. [\[ans\]](#)
5. Ask for a number and print its multiplication table up to 10 (e.g., $5 \times 1 = 5$, $5 \times 2 = 10$, ...).
6. Print all even numbers from 1 to N, then print all odd numbers from 1 to N.
7. Calculate the sum of the digits of a given number (e.g., 123 -> $1+2+3=6$). [\[ans\]](#)
8. Find the largest digit in a given integer (e.g., 48291 -> 9).
9. Determine if a given number is a prime number using a loop.
10. Print the first N terms of the Fibonacci series (0, 1, 1, 2, 3, 5, ...).
11. Print all prime numbers between 1 and a given number N using nested loops.
12. Find and print all Armstrong numbers between 1 and N. An Armstrong number is a number that is equal to the sum of the cubes of its own digits (e.g., $153 = 1^3 + 5^3 + 3^3$). [\[ans\]](#)
13. Find and print all prime numbers between 1 and 100.
14. Find the sum of digits for each number from 1 to N.
15. Find the GCD of two numbers using a loop (Euclidean algorithm). [\[ans\]](#)
16. Find the sum of all even digits and all odd digits separately in a given number.
17. Check if a number is a palindrome (reads the same forwards and backward, e.g., 121).
18. Calculate the power of a number without using Math.pow(). Take base and exponent as input.
19. Count the total number of digits in a given integer. [\[ans\]](#)
20. Print all the alphabets from 'A' to 'Z' using a for loop.
21. Calculate the sum of the series. Series ($1+2+3+\dots+N$)
22. Calculate the sum of the squares of the first N natural numbers. Series ($1^2 + 2^2 + \dots + N^2$)
23. Find the frequency of a specific digit in a given number.
24. Find the LCM of two numbers. (hint: $a*b/\text{gcd}(a,b)$) [\[ans of gcd\]](#)
25. Sum of Prime Numbers in a Range: Calculate the sum of all prime numbers between 1 and N.
26. Convert a binary number (given as an integer) to its decimal equivalent. [\[ans\]](#)
27. Convert a decimal number to its binary equivalent. [\[ans\]](#)

28. Find and print all factors of a given number.

29. Check if a number is a perfect number. A perfect number is a positive integer that is equal to the sum of its proper positive divisors (e.g., $6 = 1+2+3$).

30. Check if a number is a strong number. A number is strong if the sum of the factorial of its digits is equal to the number itself (e.g., $145 = 1! + 4! + 5!$). [\[ans\]](#)

31. Calculate the sum of the series: $1 - 2 + 3 - 4 + 5 \dots N$.

32. Find a Number and its Reverse's Sum: Take a number, reverse it, and print the sum of the original number and its reversed number.

33. Print Pascal's Triangle: Print Pascal's triangle for N rows.

34. Check if a number is a Harshad number (it is divisible by the sum of its digits).

35. Check if a number is an Automorphic number (its square ends in the same digits as the number itself, e.g., $25^2=625$).

36. Find the first and last digit of a number: Write a program to find the first and last digit of any given number.

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